

Chapter 10 Thermal Physics

Heat

The exchange of energy between objects because of temperature differences is called **heat**.

Objects are in **thermal contact** if energy can be exchanged between them.

Thermal equilibrium exists when two objects in thermal contact with each other **cease to exchange energy**.

Zeroth Law of Thermodynamics

If objects A and B are **separately in thermal equilibrium** with a third object, then A and B are **in thermal equilibrium** with each other.

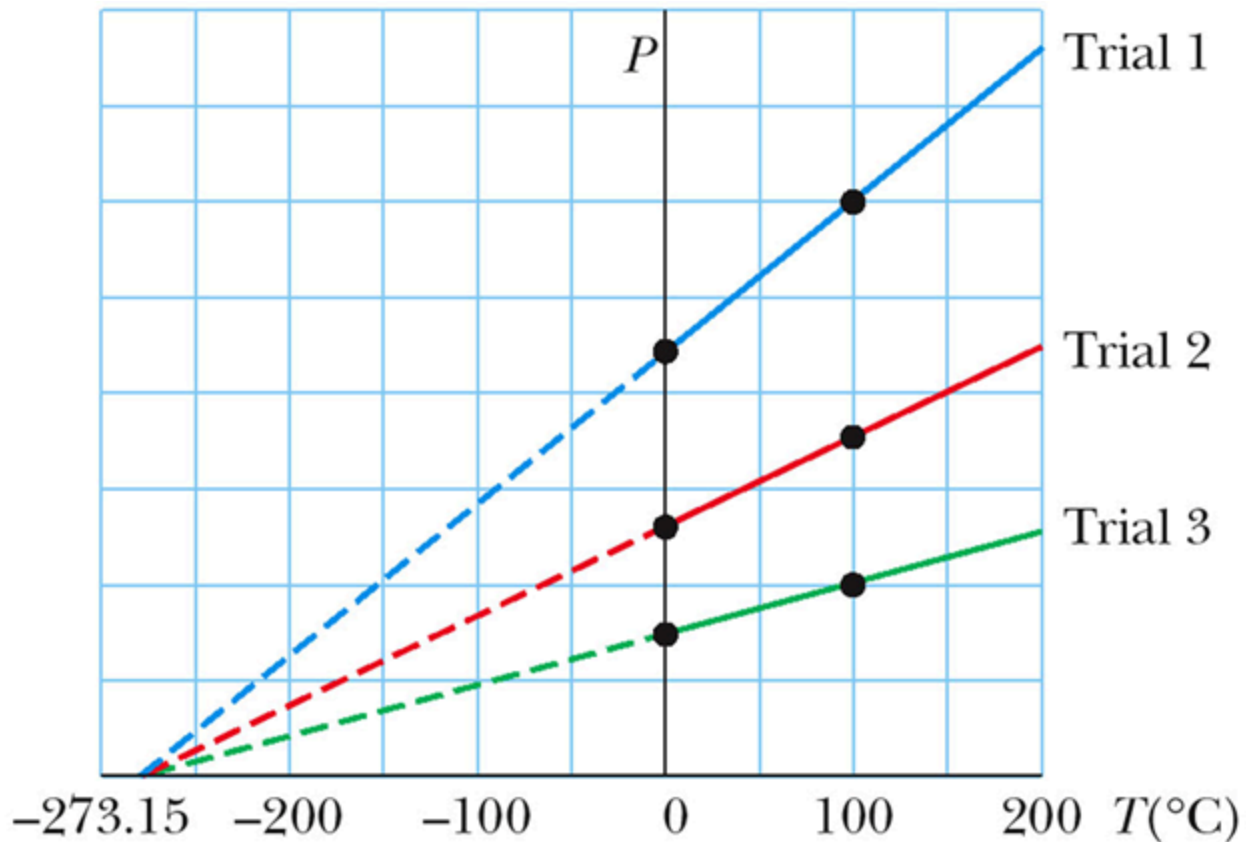
This law allows the definition of temperature.

Temperature is the property that determines **whether an object is in thermal equilibrium with other objects**.

Absolute Zero

It is obtained by **extrapolating** the law followed by ideal gases. When the temperature drops to -273.15°C , the **volume or pressure** of the gas will decrease to zero.

Absolute zero has never been reached.



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Modern Definition of Kelvin Scale

It is defined in terms of **two points**, one of which is **absolute zero** and the second is the **triple point of water** (single point where water can exist as solid, liquid and gas at the same time). It is 273.16K .

Superconductivity

The phenomenon that the electrical resistance of certain metals, alloys, and compounds suddenly decreases when temperature drops to **near absolute zero**.

Converting Among Temperature Scales

$$T_c = T_K - 273.15$$

$$T_F = \frac{9}{5}T_c + 32$$

$$T_C = \frac{5}{9}(T_F - 32)$$

$$\Delta T_F = \frac{9}{5}\Delta T_C$$

Thermal Expansion

Thermal expansion is the consequence of the change in **average separation** between its constituent atoms or molecules. As temperature increases, the **vibrating amplitude increases**. And it causes the overall object to expand.

Structure of Graphite

The structure **between layers** exhibits the characteristics of **atomic crystal**, and the structure **in the layer** exhibits the characteristics of **molecular crystal**.

Linear Expansion

For small change in temperature,

$$\Delta L = \alpha L_0 \Delta T$$

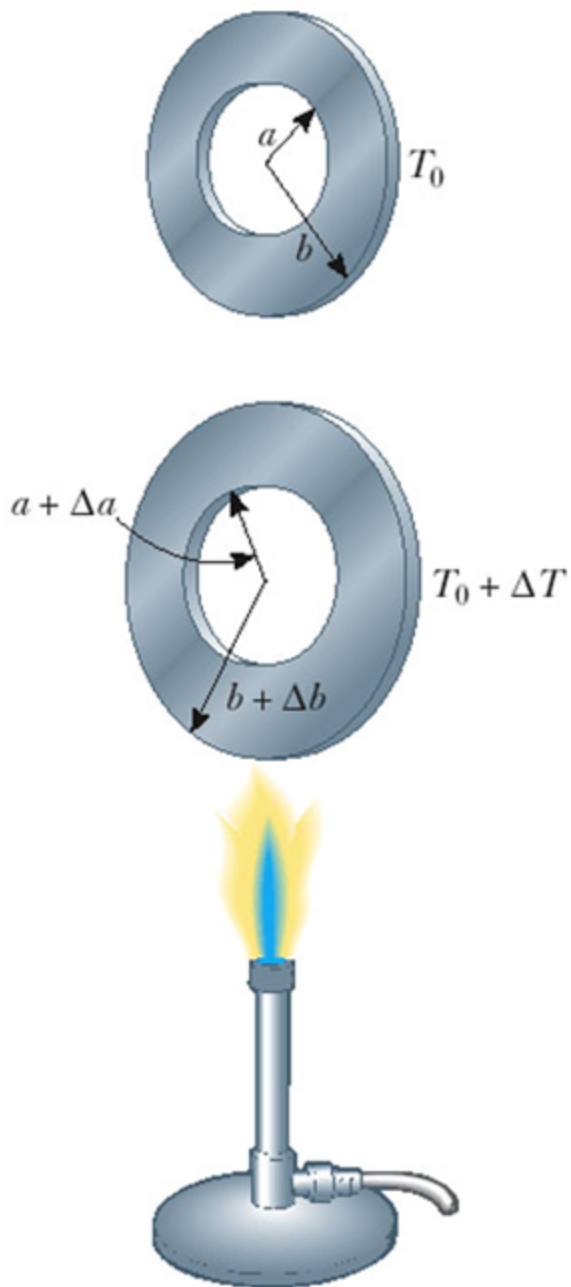
where α is the **coefficient of linear expansion**, depends on the material(average coefficients).

Area Expansion

Two dimensions expand according to

$$\Delta A = \gamma A_0 \Delta t, \gamma = 2\alpha$$

where γ is the coefficient of area expansion.



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Volume Expansion

Three dimensions expand

$$\Delta V = \beta V_0 \Delta t$$

for solids, $\beta = 3\alpha$.

Unusual Behavior of Water

When temperature of water increases from 0 to 4°C, its density increases.

Maximum density of water is 1000 kg/m^3 at 4°C.

Heat

The transfer of **energy** between a system and its environment because of **temperature difference**.

And Q is used to represent the amount.

The unit is cal. A calorie is the amount of energy to raise the temperature of 1g water from 14.5 to 15.5 °C. And $1\text{cal} = 4.186\text{J}$, which is called the **mechanical equivalent of heat**.

Specific Heat

The specific heat, c , is the requirement of amount of **energy per unit mass** to change **the temperature** of that substance by 1°C.

$$c = \frac{Q}{m\Delta T}$$

where ΔT is still **the final temperature minus the initial temperature**. If ΔT is positive then ΔQ is also positive. The energy flows into the system. And vice versa.

Calorimetry

Calorimetry is the analysis performed using a **calorimeter**, which can reach thermal equilibrium without loss energy to environment.

Conservation of energy in a isolated system:

$$Q_{\text{cold}} = -Q_{\text{hot}}$$

Calorimeter

A device to measure the **calorific value** of solids and liquids.

The **heat capacity** is the **amount of heat absorbed per 1°C**.

$$C = \frac{Q}{\Delta T}$$

Phase Changes

When the physical characteristics of the substance change from one to another such as solid to liquid.

Phases changes involve a change in the **internal energy** but **no change in temperature**.

Latent Heat

The absorbed or emitted heat during a phase change.

$$Q = \pm mL$$

where L is the latent heat of the substance, which depends on substance and nature of phase change.

Latent heat of fusion L_f is used for melting or freezing. Latent heat of vaporization L_v is used for boiling or condensing.

Sublimation

The process that some substance go directly from solid to gas without passing liquid phase. Latent heat of sublimation is associated with the change.

Affection Factors

- Intermolecular Forces:
Interactions weaken, allowing molecules to gain energy to directly from solid to gas.
- Thermodynamic Equilibrium:
The chemical potential difference between the solid and gas phases is not sufficiently large.
- Kinetic Factors:
The diffusion rate of molecules and the rate of surface reconstruction